

Linear Search

Session 3

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Outline

- 1 Beats and Bytes
- 2 Linear Search: Concept and Analogy
- 3 Coding and Predictions
- 4 Analysis

Beats and Bytes

“The successful warrior is the average man, with laser-like focus.”

Linear Search: Concept and Analogy

How do you usually keep your room?

LEFT SIDE



RIGHT SIDE



Searching & Sorting: 2 sides of the same coin

- Phone contact list
- YouTube Video
- Instagram Feed

Guess a number game!

Animation for linear search

Logic

- ➊ Visit each element one by one
 - ➊ Check element at index 0
 - ➋ Check element at index 1
 - ➌ Check element at index i . If any element matches the required value, print its index and STOP
 - ➍ ...
 - ➎ ...
 - ➏ Check element at index $n-1$
- ➋ If you have checked all elements and still haven't found the item, print -1

The FLAG method

```
flag ← false
```

```
for i ← 0 to n-1 do  
    if A[i] == key then  
        print "Element found at index", i  
        flag ← true  
        break  
    end if  
end for
```

```
if flag == false then  
    print "-1"  
end if
```

Logic

- ➊ Visit each element one by one
 - ➊ Check element at index 0
 - ➋ Check element at index 1
 - ➌ Check element at index i . If any element matches the required value, print its index and STOP
 - ➍ ...
 - ➎ ...
 - ➏ Check element at index $n-1$
- ➋ If you have checked all elements and still haven't found the item, print -1

The BOUNDARY CHECK Method

```
for i ← 0 to n-1 do
    if A[i] == key then
        print "Element found at index", i
        break
    end if
end for

if i == n then
    print "-1"
end if
```

Logic

1 Visit each element one by one

1 Check element at index 0

2 Check element at index 1

3 Check element at index i . If it matches the required value, print its index and STOP

4 ...

5 ...

6 Check element at index $n-1$

2 If you have checked all elements and still haven't found the item, print -1

for ($i=0; i < n; i++$)

if ($A[i] == key$)

print(i)
return

print(-1) return

The EARLY EXIT Method (Experience speaks)

```

for i ← 0 to n-1 do
  if A[i] == key then
    print "Element found at index", i
    return
  end if
end for

print "-1"

```

Index

$n = 5$

23	34	14	16	18
0	1	2	3	4

key = 16

key = 100

El found at index 3

0-2-3

0-2-3

4-1

Coding and Predictions

How many steps will it take?

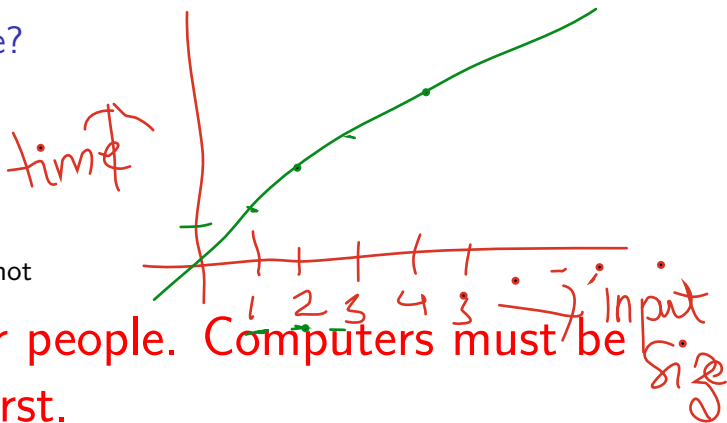
- Lucky $\rightarrow$ Best case $\rightarrow$
- Unlucky $\rightarrow$ Worst $\rightarrow$
- Sometimes lucky, sometimes not $\rightarrow$ Average

$$\begin{matrix} \sqrt{n} \\ n \\ n/2 \end{matrix}$$

Analysis

How many steps will it take?

- Lucky
- Unlucky
- Sometimes lucky, sometimes not



Optimism is fine for people. Computers must be prepared for the worst.

Flash Debate

“If I search carefully, I can find the word “rabbit” just as fast in a randomly arranged book as in an alphabetically arranged book.”